

При помощи экземпляра класса `concurrency::combinable` ускорить вычисление интегралов по квадратурной формуле Симпсона

$$\int_a^b f(x)dx \approx \frac{h}{3} \left[f(a) + f(b) + 4 \sum_{k=1}^N f(a + (2k-1)h) + 2 \sum_{k=1}^{N-1} f(a + 2kh) \right]$$

$$h = (b - a) / (2N), \quad N \gg 1$$

Вычислить $F(y) = \int_0^\infty f(x, y) dx$

$$1. \quad f(x, y) = \frac{1}{e^{-\sqrt{|x|}} + e^{\sqrt{|x|}}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{y^2 + x^2 + e^{\sqrt{k}}} \right)$$

$$2. \quad f(x, y) = e^{-\sqrt{|x|}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\sqrt{y^2 + x^2 + e^{\sqrt{k}}}} \right)$$

$$3. \quad f(x, y) = \frac{1}{e^{-\sqrt{|x|}} + e^{\sqrt{|x|}}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\sqrt{y^2 + x^2 + e^{\sqrt{k}}}} \right)$$

$$4. \quad f(x, y) = e^{-\sqrt{|x|}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\sin^2 y + e^{\sqrt{k}}} \right)$$

$$5. \quad f(x, y) = e^{-\sqrt{|x|}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\cos^2 y + e^{\sqrt{k}}} \right)$$

$$6. \quad f(x, y) = e^{-\sqrt{|x|}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\sin^2(x + y) + e^{\sqrt{k}}} \right)$$

$$7. \quad f(x, y) = e^{-\sqrt{|x|}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\cos^2(x + y) + e^{\sqrt{k}}} \right)$$

$$8. \quad f(x, y) = \frac{1}{e^{-\sqrt{|x|}} + e^{\sqrt{|x|}}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{y^2 + x^2 + e^{\sqrt{k}} \cos \frac{1}{k+1}} \right)$$

$$9. \quad f(x, y) = e^{-\sqrt{|x|}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\sqrt{y^2 + x^2 + e^{\sqrt{k}} \cos \frac{1}{k+1}}} \right)$$

$$10. \quad f(x, y) = \frac{1}{e^{-\sqrt{|x|}} + e^{\sqrt{|x|}}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\sqrt{y^2 + x^2 + e^{\sqrt{k}} \cos \frac{1}{k+1}}} \right)$$

$$11. \quad f(x, y) = e^{-\sqrt{|x|}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\sin^2 y + e^{\sqrt{k}} \cos \frac{1}{k+1}} \right)$$

$$12. f(x, y) = \frac{1}{e^{-\sqrt{|x|}} + e^{\sqrt{|x|}}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{y^2 + x^2 + e^{\sqrt{k}} \sin \frac{k+1}{k+2}} \right)$$

$$13. f(x, y) = e^{-\sqrt{|x|}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\sqrt{y^2 + x^2} + e^{\sqrt{k}} \sin \frac{k+1}{k+2}} \right)$$

$$14. f(x, y) = \frac{1}{e^{-\sqrt{|x|}} + e^{\sqrt{|x|}}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\sqrt{y^2 + x^2} + e^{\sqrt{k}} \sin \frac{k+1}{k+2}} \right)$$

$$15. f(x, y) = e^{-\sqrt{|x|}} \prod_{k=0}^{\infty} \cos \left(\frac{x}{\sin^2 y + e^{\sqrt{k}} \sin \frac{k+1}{k+2}} \right)$$